# Take-Home Assignment weeks 37–38

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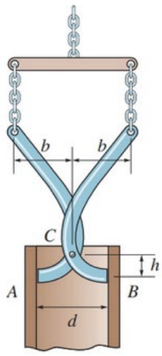
## Exercise P5-1

The pipe is hoisted using the tongs. If the coefficient of static friction at $A$ and $B$ is $\mu_s$, determine the smallest dimension $b$ so that any pipe of inner diameter $d$ can be lifted.



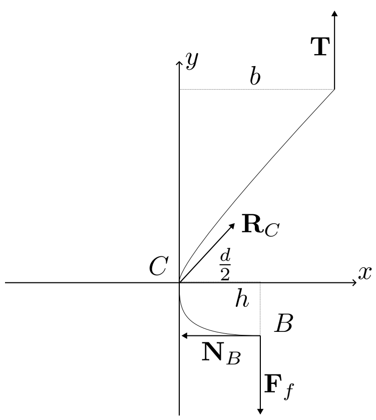




Figure 0.1: Free-body diagram of the right tong half

We consider one half of the tong, in this case the right half as shown on Figure 0.1. For this to be in equilibrium both the vector sum of all forces and moments about any point must be zero. I.e.

$$\mathbf{F}_R = \sum \mathbf{F} = \mathbf{0}$$
$$\mathbf{M}_O = \sum \mathbf{M}_O = \mathbf{0}.$$

For a couple moment not to arise, the lines of actions of all the forces must intersect at some common point, $P$, i.e. the forces must be concurrent. If the forces are not concurrent, there will always be some "perpendicular distance" between the lines of action of the forces and therefore a couple moment might be produced even if force equilibrium is required.

At point $B$, right at the verge of slippage, the line of action of the resultant force at $B$ (which is the sum of the friction force and normal force components at this point) makes the friction angle, $\phi = \tan^{-1}\mu_s$ below the horizontal. We can therefore find the equation of the line using the formula for a line through a point, which in this case is $\left(-h, \frac{d}{2}\right)$ and a slope, which in this case is $\mu$ as:

$$y + h = \mu_s(x - \frac{d}{2}) \implies y = \mu_s\left(x - \frac{d}{2}\right) - h.$$

The line of action at point $B$ will lie along this line.

At the top the tong is attached to a vertical chain, such that the force that is transmittet to the tong at the top is vertical, with a line of action of $x = b$.

As mentioned above, all the lines of actions of the forces must intersect at a common point $P$. This means that the two lines defined above must intersect at this point $P$. The intersection between the line of action of $\mathbf{T}$ and the line of action of $\mathbf{R}_B$ can now be found as:

$$P = \left(b, \mu_s\left(b - \frac{d}{2}\right) - h\right).$$

The pin force $\mathbf{R}_C$ at point $C$ will act along the line $CP$. If this point $P$ is below point $C$ one can imagine that the tongues will "open up" and not press against the wall. Therefore, for the jaws to wedge and hold, the concurrency point $P$ must be at or above $C$ as:

$$\begin{aligned} y_p &\geq 0 \\ \mu_s\left(b - \frac{d}{2}\right) - h &\geq 0 \\ \implies \mu_s\left(b_{\min} - \frac{d}{2}\right) - h &= 0 \\ b_{\min} &= \frac{h}{\mu_s} + \frac{d}{2}. \end{aligned}$$

Therefore the smallest dimension of $b$ such that the pipe can be lifted is $b_{\min} = \frac{h}{\mu_s} + \frac{d}{2}$.